

Alvaro Perdomo Gutierrez

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **12+** years across fintech, healthcare, and e-commerce, building production AI systems and scaling ML platforms with Python, Node.js, and cloud infrastructure. Expert in LLM orchestration, RAG pipelines, agent architectures, and vector search integration using **LangChain** and **LlamaIndex**. Proven track record leading cross-functional teams, managing complex client engagements, and delivering AI-powered systems that compress costs and automate critical workflows.

SKILLS

Languages: Python, JavaScript, TypeScript, Go, Bash, SQL, HTML, CSS

Generative AI: LangChain, LlamaIndex, OpenAI API, Anthropic API, LangGraph, CrewAI, Hugging Face, FAISS, Pinecone, ChromaDB, Weaviate, Qdrant, Ollama, vLLM, MLflow, LangSmith

Data & ML: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Spark, Airflow, dbt

Backend: Node.js, FastAPI, Django, Express, Flask, Kafka, RabbitMQ, gRPC

Frontend: React, Next.js, Vue.js, Angular, Tailwind CSS, Material-UI, Redux, Webpack

Mobile: React Native, Flutter, Expo, Swift, Kotlin

Database: PostgreSQL, MongoDB, Redis, DynamoDB, Elasticsearch, TcaplusDB, Cassandra

Testing: Jest, Pytest, Playwright, Cypress, Selenium, k6

Cloud & DevOps: AWS, GCP, Azure, Kubernetes, Docker, Terraform, GitHub Actions, Jenkins, CloudWatch, S3, EC2, RDS, Lambda, CloudFront, Heroku

Others: RESTful API, GraphQL, WebSocket, Prometheus, Grafana, Datadog, Git, GitHub, Jira, Confluence, Agile, Scrum

EXPERIENCE

Leanware

May 2024–Present

Senior Software Engineer

- Spearheaded development of an AI-powered PRD agent using **Python** and **LangChain**, orchestrating multi-step workflows that transformed product documentation from manual drafting into automated, context-aware specification generation across 6+ client sectors.
- Architected full-stack infrastructure for an AI developer-productivity platform using **TypeScript**, **React**, and **FastAPI**, integrating **GitHub API** analysis with **LlamaIndex**-powered code understanding to surface actionable insights for development teams.
- Engineered deployment pipelines across multiple concurrent engagements using **Docker**, **Kubernetes**, and **AWS**, ensuring production reliability and enabling rapid iteration on AI features while maintaining security and compliance standards required by B2B clients.

- Led architectural decisions for AI fitness application incorporating **OpenAI API** integration and **Pinecone** vector search, reducing latency on personalization queries by **62%** through optimized embedding retrieval and prompt caching strategies.
- Mentored cross-functional team on production AI system patterns: prompt engineering, evaluation frameworks using **LangSmith**, and structured output validation, establishing reusable patterns that accelerated subsequent project deliveries by an average of **3 weeks**.
- Built automated evaluation harness for AI models across the portfolio using **Pytest** and **MLflow**, creating standardized metrics and benchmarks that allowed the team to confidently iterate on model selection and fine-tuning without regression.
- Collaborated with product and strategy teams to translate vague requirements into technical specifications: decomposed complex agent use cases into discrete orchestration steps, identified feasibility concerns early, and communicated trade-offs between capability and latency.
- Developed observability infrastructure using **Prometheus** and **Grafana** for all production systems, surfacing LLM cost metrics, token usage patterns, and inference latency alongside traditional application metrics to optimize operational spend.
- Refactored legacy codebase across three engagements by replacing inconsistent error handling with structured exception patterns and adding comprehensive logging, reducing time-to-diagnosis for production incidents by **45%**.
- Balanced shipping velocity with code quality by implementing strict code review standards for **GitHub** pull requests, giving direct feedback to junior engineers on architecture patterns and edge case handling, which improved first-pass review acceptance rates by **38%**.
- Owned delivery of dental evaluation AI system end-to-end: scoped feasibility, coordinated with clinical domain experts to validate model outputs, deployed **TensorFlow**-based computer vision pipeline to **AWS**, and managed client handoff with documentation and operational runbooks.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Led integration of AI Shopping features including image-based product search and natural-language style chat using **LangChain** and **OpenAI API**, shipping multimodal capabilities that increased user engagement and contributed to **2.8%** improvement in iOS conversion rates.
- Owned migration of ML model serving infrastructure from legacy batch systems to **Databricks** unified analytics platform using **Python** and **Spark**, enabling real-time model evaluation and reducing MVP deployment time from 2 quarters to 1 month and subsequent iterations from 2 months to 2 weeks.
- Architected integration with Lily AI, processing **275K** images weekly with automated tagging and metadata enrichment, leading to **15%** increase in seller sell-through and reducing manual curation workload by eliminating **40%** of redundant review steps.
- Designed and deployed garment-measurement ML model incorporating **PyTorch** and **TensorFlow** with structured validation pipelines, serving **100K+** predictions daily with **99.2%** uptime across the resale platform supporting **172M+** items.

- Engineered personalization recommendation engine using **Redis** for session caching and **PostgreSQL** for feature materialization, optimizing single-SKU purchase probability calculation and reducing average recommendation latency from **1.2s** to **320ms**.
- Implemented automated photography and visual recognition pipeline that processed high-volume seller uploads using **Python** image processing, **Airflow** orchestration, and **AWS Lambda** for scale, enabling consistent product quality across **55K+** brands.
- Collaborated with product and data science teams to evaluate new embedding models and vector databases, conducting benchmarks using **FAISS** and **Weaviate** and recommending architecture changes that balanced search relevance against operational cost.
- Built comprehensive observability for ML systems using **Datadog**, **Prometheus**, and custom **Python** instrumentation, tracking model drift, feature staleness, and prediction confidence to maintain clinical-grade performance even as inventory volume scaled **6x** year-over-year.
- Conducted detailed code reviews for **TypeScript** and **Python** backend changes across the platform, catching edge cases in price prediction logic and SQL query optimization that would have impacted profitability and user trust.
- Addressed recurring customer experience issues by diagnosing root causes in seller processing delays: identified bottleneck in image tagging service queue, implemented backpressure handling in **RabbitMQ**, and automated retry logic that reduced complaint volume by **67%** in 2023.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Played a key role in scaling precision-matching ML platform supporting **2M+** members across **6x** growth trajectory, building **Python**-based matching algorithms that preserved clinical validity while handling increased complexity and geographic diversity.
- Engineered global Care Navigation platform expansion incorporating **LlamaIndex** for context-aware recommendation generation, enabling localized clinical guidance across multiple jurisdictions and reducing navigation friction by **22%** as measured by engagement metrics.
- Built and deployed global family-plan launch feature (September 2021) with jurisdiction-aware member matching and localized eligibility validation, supporting dependent enrollment across **15** countries with **PostgreSQL** schema updates ensuring compliance with regional regulations.
- Scaled HIPAA-compliant infrastructure to support Fortune 500 onboarding (PepsiCo, Instacart, Bain) by architecting **AWS** networking, encryption, and audit logging standards; migrated data handling to eliminate compliance gaps and enabled **4** major enterprise clients to go live ahead of schedule.
- Developed operational ML models for member retention using **scikit-learn** and **Python** including dropout-risk prediction and demand forecasting, enabling proactive outreach that improved member lifetime value and reduced churn by **18%** in the first cohort.
- Collaborated with clinical and product teams to translate mental health domain requirements into technical specifications: decomposed complex member-matching logic, identified when heuristics needed to be replaced with ML, and proposed architectural trade-offs between accuracy and latency.

- Optimized **PostgreSQL** queries for member eligibility checks and benefit lookups, reducing average response time from **850ms** to **260ms** through index optimization and query plan analysis, supporting **10x** concurrent user growth without infrastructure scaling.
- Implemented comprehensive logging and monitoring using **Datadog** and **Prometheus**, surfacing member data access patterns and system performance anomalies critical to compliance audits and operational trust during rapid scaling.

Tencent Games

Nov 2014–Nov 2020

Senior Game Engineer

- Contributed to Honor of Kings mobile MOBA infrastructure as part of TiMi Studios, architecting real-time player state synchronization using **Go** and **gRPC** that scaled from launch to **100M** daily active users by November 2020, handling **40M+** daily peak concurrent connections.
- Built anti-cheat systems including kernel-level **ACE** (Tencent Anti-Cheat Engine) components and behavioral detection using **Python** anomaly detection models, reducing reported cheating incidents by **89%** and maintaining competitive integrity across the player base.
- Engineered PUBG Mobile infrastructure component in collaboration with LIGHTSPEED and Krafton using **Unreal Engine 4**, supporting **100-player** battle royale gameplay with **<100ms** latency targeting across Asia-Pacific server clusters on **GCP** compute infrastructure.
- Developed game server backend using **C++** and **TcaplusDB** NoSQL database handling **40M+** daily transactions, optimizing data serialization and connection pooling to maintain **99.8%** uptime during peak hours while reducing per-player storage costs by **31%**.
- Collaborated on Tencent Gaming Buddy emulator (2018) providing desktop access to mobile titles, architecting compatibility layers and input translation logic in **C++** and **JavaScript** that enabled seamless cross-platform play without performance degradation.
- Implemented matchmaking and rating systems for Arena of Valor cross-platform title using **Node.js** and **PostgreSQL**, managing skill-based queue distribution across millions of concurrent players and maintaining **<90 second** queue times at **p99** across global regions.
- Contributed to FIFA Online 3 and FIFA Online 4 operations and feature releases, managing live service infrastructure for the **#1** sports game in China and supporting e-sports World Cup broadcasts through high-reliability deployment pipelines and **Kafka**-based event streaming.
- Mentored junior engineers on performance optimization and debugging distributed game systems, conducting code reviews for backend **Python** services and infrastructure code, establishing patterns for error handling and graceful degradation that became team standards.

EDUCATION

The Hong Kong Polytechnic University

Sep 2010–Nov 2014

Bachelor of Science in Computer Science